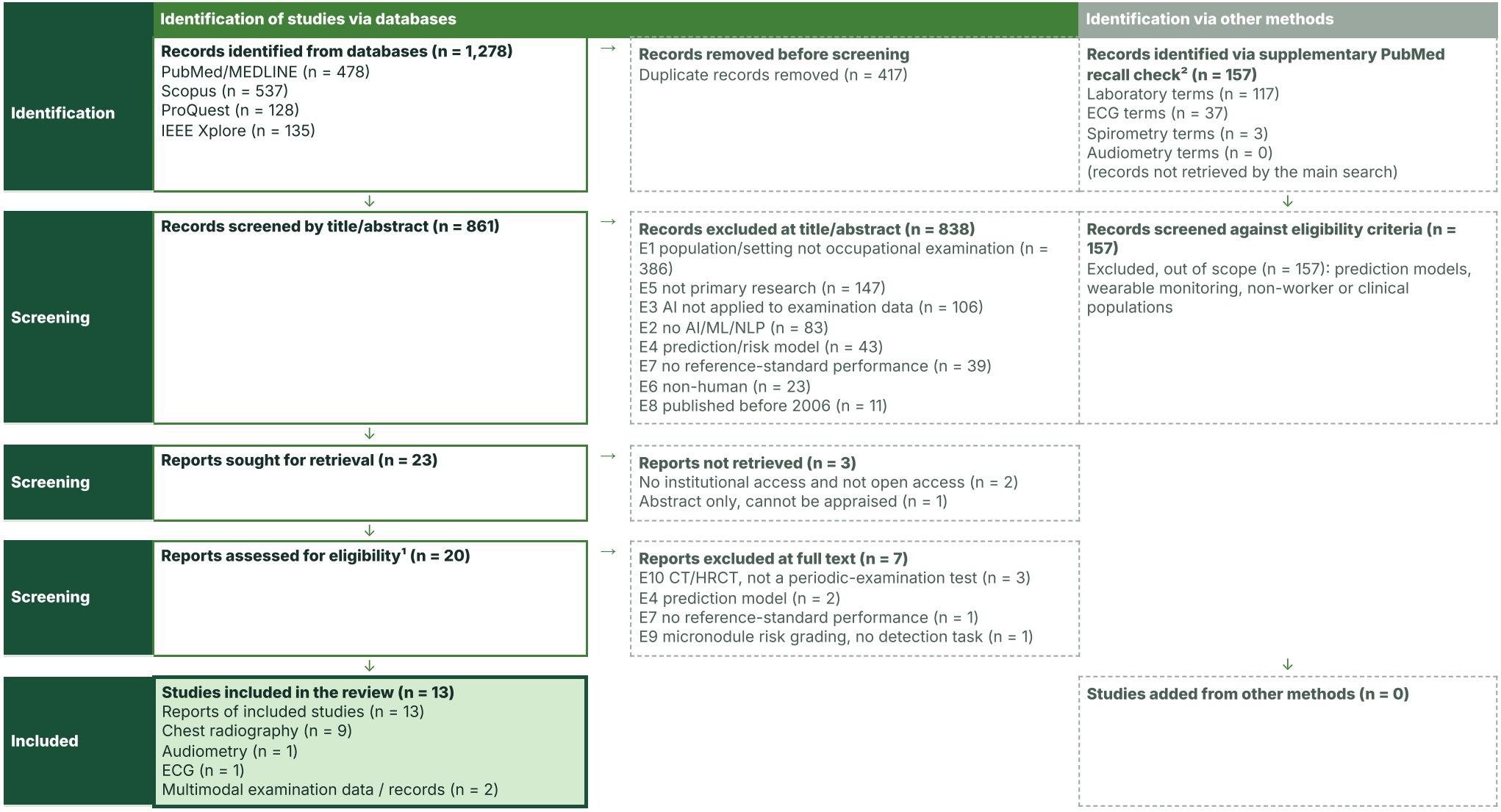
# Artificial Intelligence for Interpreting Occupational Health Examination Tests: A Systematized Review of Evidence and Gaps

Supplement to the e-poster, 1st PERDOKI Jakarta Summit 2026. Vidya Cecilia, Occupational Medicine Residency Program, Faculty of Medicine, Universitas Indonesia.

## PRISMA 2020 flow diagram




[1] Full-text eligibility (rule set 7 Oct 2026): population judged together with the index test. Workers or ex-workers examined with a test used in periodic or pre-placement occupational examinations were eligible from screening, surveillance, clinic or claims settings. E9 and E10 were defined at full text; title/abstract decisions preceded this rule.
[2] PubMed probe (6 Oct 2026): AI terms AND modality terms AND worker terms, 2006 onwards, run because laboratory and ECG terms were absent from the main strings; counts are records not already retrieved by the main search.

## Included studies (n = 13)


1. Heo SJ, Kim Y, Yun S, Lim SS, Kim J, Nam CM, et al. Deep learning algorithms with demographic information help to detect tuberculosis in chest radiographs in annual workers' health examination data. Int J Environ Res Public Health. 2019;16(2):250. doi: 10.3390/ijerph16020250
2. Wang X, Yu J, Zhu Q, Li S, Zhao Z, Yang B, et al. Potential of deep learning in assessing pneumoconiosis depicted on digital chest radiography. Occup Environ Med. 2020;77(9):597-602. doi: 10.1136/oemed-2019-106386
3. Young C, Barker S, Ehrlich R, Kistnasamy B, Yassi A. Computer-aided detection for tuberculosis and silicosis in chest radiographs of gold miners of South Africa. Int J Tuberc Lung Dis. 2020;24(4):444-51. doi: 10.5588/ijtld.19.0624
4. Charapaqui-Miranda S, Arapa-Apaza K, Meza-Rodriguez M, Chacon-Torrico H. Comparing predictive machine learning algorithms in fit for work occupational health assessments. In: Lossio-Ventura JA, Condori-Fernandez N, Valverde-Rebaza JC, editors. Information management and big data. 6th International Conference, SIMBig 2019; 2019 Aug 21-23; Lima, Peru. Cham: Springer; 2020. p. 218-25. (Communications in computer and information science; vol. 1070). doi: 10.1007/978-3-030-46140-9_21
5. Ehrlich R, Barker S, Te Water Naude J, Rees D, Kistnasamy B, Naidoo J, et al. Accuracy of computer-aided detection of occupational lung disease: silicosis and pulmonary tuberculosis in ex-miners from the South African gold mines. Int J Environ Res Public Health. 2022;19(19):12402. doi: 10.3390/ijerph191912402
6. Wang Y, Chen Z, Tian S, Zhou S, Wang X, Xue L, et al. Convolutional neural network-based ECG-assisted diagnosis for coal workers. Int J Environ Res Public Health. 2022;20(1):9. doi: 10.3390/ijerph20010009
7. Li X, Xu M, Yan Z, Xia F, Li S, Zhang Y, et al. Deep convolutional network-based chest radiographs screening model for pneumoconiosis. Front Med (Lausanne). 2024;11:1290729. doi: 10.3389/fmed.2024.1290729
8. Liu M, Huang Z, Borek M, Loveless I, Rosenman K, Alessio A, et al. Pneumoconiosis screening and classification using deep learning models. Occup Environ Med. 2026;83(5):276-82. doi: 10.1136/oemed-2026-111040
9. He L, Luo Y, Lei J, Yin Q, Yang D, Zhang C, et al. [Evaluation of clinical value of interpretable Transformer-based models in computer-aided diagnosis of early-stage pneumoconiosis]. J Environ Occup Med. 2026;43(7):844-51. Chinese. doi: 10.11836/JEOM25477
10. Vella F, Senia P, Filetti V, Vitale E, Sferrazzo G, Privitera L, et al. Application of large language models as decision support tools in occupational health and safety management: a cohort study of industrial workers. Front Public Health. 2026;14:1815316. doi: 10.3389/fpubh.2026.1815316
11. Moore BCJ, Schlittenlacher J. Diagnosing noise-induced hearing loss sustained during military service using deep neural networks. Trends Hear. 2023;27:23312165231184982. doi: 10.1177/23312165231184982
12. Priego-Torres B, Sanchez-Morillo D, Khalili E, Conde-Sánchez MÁ, García-Gámez A, León-Jiménez A. Automated engineered-stone silicosis screening and staging using deep learning with X-rays. Comput Biol Med. 2025;191:110153. doi: 10.1016/j.compbiomed.2025.110153
13. Dong H, Zhu B, Zhang X, Kong X. Use data augmentation for a deep learning classification model with chest X-ray clinical imaging featuring coal workers' pneumoconiosis. BMC Pulm Med. 2022;22(1):271. doi: 10.1186/s12890-022-02068-x

## Study characteristics and main results

| # | Study | Country | Modality / target | Reference standard | Main results |
|---|---|---|---|---|---|
| 1 | Heo 2019 (PMID 30654560) | South Korea | CXR / tuberculosis | Radiologist reading under the national QA programme | Test AUC VGG19 0.9075 (image-only) to 0.9213 (+ demographics), p < 0.05 |
| 2 | Wang 2020 (PMID 32471837) | China | CXR / pneumoconiosis vs normal | Expert panel (1 certified radiologist + 4 occupational physicians), ILO | AUC 0.878 (95% CI 0.811-0.946) vs radiologists 0.668 and 0.772 |
| 3 | Young 2020 (PMID 32317070) | South Africa | CXR / silicosis, TB, silicotuberculosis | Radiologist + 3 physicians (board); 4 expert readers (screening set) | Best system AUC 0.986 for any abnormality (Se 98.2, Sp 98.2); TB vs normal 0.989; silicosis vs normal 0.986 (system A), 0.939 (B) |
| 4 | Charapaqui-Miranda 2020 (Scopus 85084850887) | Peru | Multimodal exam data / work-entry MCU fit/unfit category (fit or fit with modifications vs unfit) | Fit/unfit result recorded by the examining doctor (MCU status, not a 7-step fitness-to-work assessment) | Best model logistic regression: AUC 60.4%, accuracy 68.1% |
| 5 | Ehrlich 2022 (PMID 36231700) | South Africa | CXR / silicosis, TB, any abnormality | 2 occupational medicine specialists (reader 2 = consensus of two) | Silicosis AUC 0.883 and 0.839 (ILO >= 1/0), 0.887 and 0.892 (>= 1/1) vs reader 1 and 2 (system A) |
| 6 | Wang 2022/2023 (PMID 36612331) | China | 12-lead ECG / 4 abnormal ECG classes | ECG diagnosis from the examination (cardiologist-performed ECG; labelling procedure not stated explicitly) | Test accuracy 93.0-97.7%, AUC 0.92-0.98 across 4 ECG classes |
| 7 | Li 2024 (PMID 38348336) | China | CXR / pneumoconiosis vs normal | 10 occupational disease experts + 2 radiologists, 3-5 labellers per film | Validation set (261 pneumoconiosis, 234 non): accuracy 95%, AUC 0.947 |
| 8 | Liu 2026 (PMID 42457633) | USA | CXR / 4 ILO-aligned tasks (1-2 screening, 3-4 grading) | Consensus of certified B-readers (2 + third on disagreement) | In-distribution accuracy 0.93, 0.92, 0.91 (tasks 1-3), 0.83 (task 4) vs B-readers 0.90, 0.91, 0.77, 0.86 |
| 9 | He 2026 (Scopus 105046943496) | China | CXR / early pneumoconiosis, binary | Consensus of 3 senior radiologists | CrossFormer AUC 0.962 (95% CI 0.945-0.979), accuracy 92.0%, F1 0.916 |
| 10 | Vella 2026 (PMID 42180469) | Italy | Occupational health records / risk assessment, surveillance protocol, fitness judgements from documents only (4 categories; not a 7-step fitness-to-work assessment) | Experienced occupational physician | Perfect concordance of document-based fitness judgments between LLM and physician |
| 11 | Moore 2023 (PMID 37550005) | UK | Pure-tone audiometry 1-8 kHz both ears + age / military noise-induced hearing loss | Group membership: claimant with medical report excluding other causes vs unexposed control (no individual diagnosis) | MLP(18) on test databases: Se 98.6, Sp 90.2, d' 3.49 (rule-based methods Sp 40-64) |
| 12 | Priego-Torres 2025 (PMID 40252290) | Spain | CXR / silicosis screening (healthy vs silicosis) and staging (SS vs PMF) | Clinical diagnosis: exposure history, ILO reading and HRCT (ICOERD), some biopsy; controls labelled normal in VinDr | Screening AUC 1.00, accuracy 0.97-0.99 (4 CNNs); staging SS vs PMF accuracy 0.81-0.84, AUC 0.91-0.94 |
| 13 | Dong 2022 (PMID 35840945) | China | CXR / CWP radiographic features (nodules, interstitial change, emphysema) | Reports from 3 radiologists, reviewed by 2 senior radiologists | Average AUC 0.98 (nodules 0.97, interstitial 0.96, emphysema 1.00); accuracy 0.96 |

## JBI diagnostic test accuracy checklist (item level; no overall score)

| # | Study | Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 | Q9 | Q10 | Yes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Heo 2019 | Y | Y | U | Y | N | U | Y | Y | Y | U | 6/10 |
| 2 | Wang 2020 | U | N | U | Y | U | Y | Y | Y | Y | U | 5/10 |
| 3 | Young 2020 | N | N | N | Y | N | Y | Y | Y | Y | U | 5/10 |
| 4 | Charapaqui-Miranda 2020 | U | Y | N | Y | NA | U | Y | Y | Y | N | 5/10 |
| 5 | Ehrlich 2022 | Y | Y | Y | Y | Y | U | Y | Y | Y | Y | 9/10 |
| 6 | Wang 2023 (ECG) | U | Y | U | Y | U | U | U | Y | U | U | 3/10 |
| 7 | Li 2024 | N | N | U | Y | U | U | U | Y | Y | Y | 4/10 |
| 8 | Liu 2026 | N | U | U | Y | U | Y | Y | Y | Y | Y | 6/10 |
| 9 | He 2026 | N | N | N | Y | U | Y | Y | Y | Y | Y | 6/10 |
| 10 | Vella 2026 (LLM) | U | Y | U | U | NA | U | Y | Y | Y | Y | 5/10 |
| 11 | Moore 2023 (audiometry) | N | N | U | Y | U | N | Y | Y | N | Y | 4/10 |
| 12 | Priego-Torres 2025 | N | N | U | Y | NA | Y | Y | U | N | Y | 4/10 |
| 13 | Dong 2022 | N | N | N | N | NA | Y | Y | Y | Y | U | 4/10 |

Y yes · N no · U unclear · NA not applicable. Q1 consecutive or random sample; Q2 case-control design avoided; Q3 inappropriate exclusions avoided; Q4 index test read without knowledge of the reference; Q5 threshold pre-specified; Q6 reference standard likely correct; Q7 reference read without knowledge of the index test; Q8 appropriate interval; Q9 same reference for all; Q10 all participants analysed.